THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Electrical Engineering & Computer Science
Semester One Examinations, 2024
COMP4702 / COMP7703 Machine Learning

This paper is for St Lucia Campus students.

Examination Duration: 180 minutes

Planning Time: 10 minutes

Exam Conditions:

•No written or printed material permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x Gradescope Bubble Sheet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer Part A in the Gradscope Bubble Sheet. Answer Part B in the space provided in the question paper.

For Examiner Use Only

Question	Mark

Total _____

Part A - Multiple choice questions (2 marks each). Record your answers on the bubble sheet provided.

Question 1.

Which of the following is an *incorrect* statement about machine learning?

(a) Machine learning uses a set of data to determine suitable parameters for a mathematical model.

(b) Supervised learning means that the machine learning model uses the previous error of the model to predict the future error.

(c) The goal of machine learning is to achieve good generalisation performance.

(d) In machine learning, the training algorithm typically solves an optimisation problem.

Question 2.

In the k-nearest neighbours method (k-NN), k refers to:

(a) The maximum number of training iterations.

(b) The number of data points that are used to determine any prediction made by the model.

(c) The learning rate parameter.

(d) The number of base models that are combined to produce the final output.

Question 3.

Comparing linear regression and logistic regression, which of the following statements is *incorrect*?

(a) For logistic regression, training can be formulated as maximizing a likelihood function, whereas this is not possible for linear regression.

(b) Training a linear regression model can be formulated such that a closed-form solution is available, while logistic regression requires an iterative training algorithm.

(c) Linear regression is appropriate for solving regression problems, whereas logistic regression is appropriate for solving classification problems.

(d) Given p input variables and a single target variable, the standard linear and logistic regression models have the same number of parameters.

Question 4.

You are given 1000 data points to build a classifier. You shuffle the points randomly and then decide to put 10% of the data aside as a (final) test set and 20% of the data aside as a "validation" set, which you use to perform model selection and configuration (e.g. hyperparameter tuning). You then use 5-fold cross-validation training with the remaining data. How many data points are in each fold?

(a) 100.

(b) 700.

(c) 300.

(d) 140.

Question 5.

Consider training a machine learning model using gradient descent. If the value of the learning rate parameter is not suitable, possible consequences are:

(a) The cost function value decreases very slowly over iterations.

(b) The cost function value rapidly increases over iterations.

(c) The cost function value behaves randomly, jumping up and down with different values.

(d) All of the above.

Question 6.

A linear regression model is trained by solving the following optimisation problem:

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{n}||\mathbf{X}\theta - \mathbf{y}||_2^2 + \lambda||\theta||_2^2$$

In this equation, the regularisation hyperparameter is denoted by:

(a) θ.

(b) n.

(c) 2.

(d) λ.

Question 7.

Regarding Convolutional Neural Networks (CNNs), which of the following statement is *incorrect*?

(a) CNNs use techniques such as stride and pooling to reduce the overall size of the network.

(b) A CNN is only considered to be a deep neural network if the number of layers is greater than the number of hidden units in the largest layer.

(c) CNNs enforce sparse interactions by only having incoming connections to a hidden unit from a small, localized region of the previous layer.

(d) CNNs implement parameter sharing by having the same weights for all hidden units in a layer, using a filter that moves across all positions on the input.

Question 8.

An important difference between bagging and boosting is that:

(a) Bagging is suitable for regression problems, while boosting is suitable for classification problems.

(b) Bagging can build a heterogeneous ensemble of models, while for boosting the ensemble must be homogeneous.

(c) Boosting trains an ensemble of models sequentially, while bagging can be implemented in parallel.

(d) Boosting allows for ensembles of any size, while in bagging the number of base models must be less than the number of data points.

Question 9.

Regarding kernel ridge regression, which of the following statements is *incorrect*?

(a) kernel ridge regression incorporates L^2 regularisation as part of the method.

(b) kernel ridge regression uses a kernel function to compute a (dis)similarity between training points.

(c) The size of the model and computational requirements scale primarily with the number of training points, rather than the dimensionality of the input data.

(d) kernel ridge regression is typically trained using stochastic gradient descent.

Question 10.

Which of the following is *not* a generative machine learning model?

(a) k-nearest neighbour.

(b) Gaussian Process.

(c) Generative adversarial network.

(d) Gaussian mixture model.

(next question over)

Part B - Questions carry the number of marks indicated. Write your answers in the space provided after each part of each question.

Question 1.

Figure 1 shows a set of training data for a binary classification problem, where a point labelled '+' is a family car, and a point labelled '-' is not a family car.

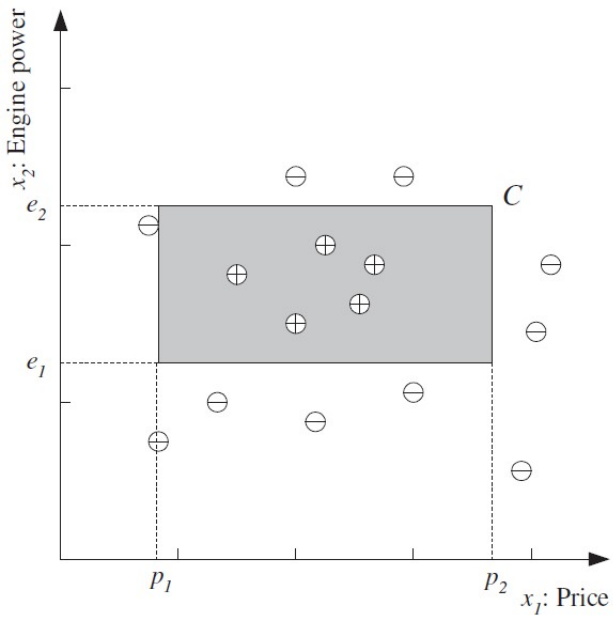

Figure 1

(a) A decision tree is trained on this data and creates the discriminant function/decision boundary shown by the shaded rectangle, C. Draw a diagram of this decision tree, including the rule/decision made at each node and the final classification at each leaf node. (4 marks)

(question continues over)

(b) If a logistic regression model was trained on this data, the best performance it could achieve would be a misclassification rate of $\frac{3}{15} = 20\%$. Draw approximately where this discriminant would be located in Figure 1. (2 marks)

(c) Consider using k-nearest neighbour to classify the data in Figure 1. If a test point $\mathbf{x}^\star$ was located at the centre of the shaded rectangle, describe how the prediction would change for different possible k values. (4 marks)

(next question over)

Question 2.

Figure 2 shows two related Figures from a machine learning textbook:

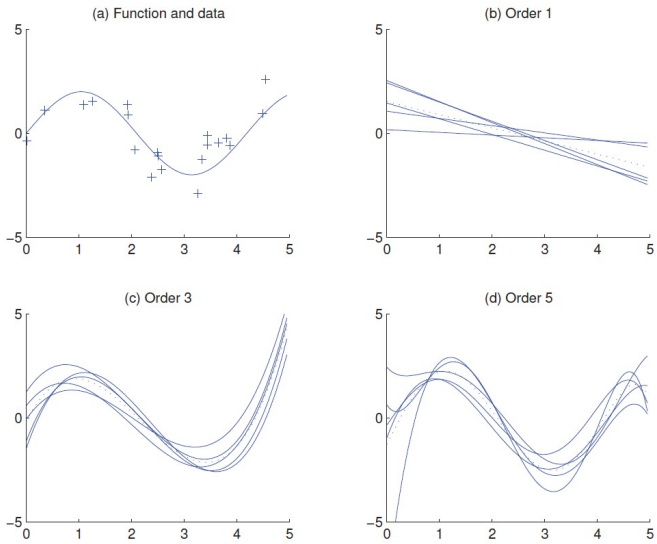

Figure 4.5 (a) Function, $f(x) = 2\sin(1.5x)$, and one noisy ($\mathcal{N}(0,1)$) dataset sampled from the function. Five samples are taken, each containing twenty instances. (b), (c), (d) are five polynomial fits, namely, $g_i(\cdot)$, of order 1, 3, and 5. For each case, dotted line is the average of the five fits, namely, $\overline{g}(\cdot)$.

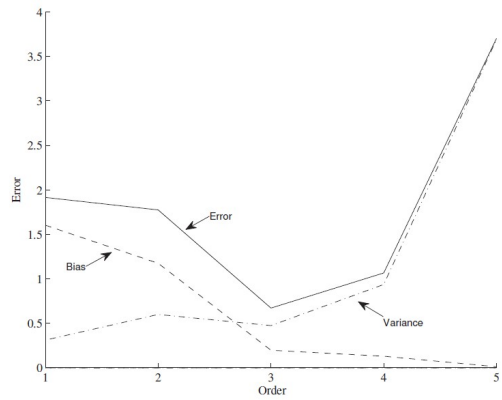

Figure 4.6 In the same setting as that of figure 4.5, using one hundred models instead of five, bias, variance, and error for polynomials of order 1 to 5. Order 1 has the smallest variance. Order 5 has the smallest bias. As the order is increased, bias decreases but variance increases. Order 3 has the minimum error.

Figure 2

(a) The caption on Fig.4.6 states that the plot was produced using 100 models (each based on a sample of twenty instances as mentioned in the caption for Fig.4.5). Explain exactly how each of the curves could have been calculated using the 100 models and sample data sets. (4 marks)

(question continues over)

(b) Explain the general relationship between underfitting, overfitting and model complexity in supervised learning. You should refer to Figure 2 to assist with your explanation. (6 marks)

(next question over)

Question 3.

Figure 3 is taken from the course textbook (Algorithm 6.1):

Algorithm 6.1: Backpropagation

Input: Parameters $\boldsymbol{\theta} = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^{L}$, activation function h, and data $\mathbf{X}$, $\mathbf{y}$, with n_b rows, where each row corresponds to one data point in the current mini-batch.

Result: $J(\boldsymbol{\theta})$, $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$ of the current mini-batch.

Forward propagation
Set $\mathbf{Q}^0 \leftarrow \mathbf{X}$
for $l = 1, \ldots, L$ **do**
  $\mathbf{Z}^{(l)} = \mathbf{Q}^{(l-1)}\mathbf{W}^{(l)\mathsf{T}} + \mathbf{b}^{(l)\mathsf{T}}$
  $\mathbf{Q}^{(l)} = h(\mathbf{Z}^{(l)})$ *Do not execute this line for last layer $l = L$*
end
Evaluate cost function
if *Regression problem* **then**
  $J(\boldsymbol{\theta}) = \frac{1}{n_b} \sum_{i=1}^{n_b} \left(y_i - Z_i^{(L)}\right)^2$
  $d\mathbf{Z}^{(L)} = -2\left(\mathbf{y} - \mathbf{Z}^{(L)}\right)$
else if *Classification problem*[2] **then**
  $J(\boldsymbol{\theta}) = \frac{1}{n_b} \sum_{i=1}^{n_b} \left(-Z_{i,y_i}^{(L)} + \ln\left(\sum_{j=1}^{M} \exp\left(Z_{ij}^{(L)}\right)\right)\right)$
  $dZ_{ij}^{(L)} = -\mathbb{I}\{y_i = j\} + \frac{\exp\left(Z_{ij}^{(L)}\right)}{\sum_{j=1}^{M} \exp\left(Z_{ij}^{(L)}\right)} \qquad \forall i, j$
Backward propagation
for $l = L, \ldots, 1$ **do**
  $d\mathbf{Z}^{(l)} = d\mathbf{Q}^{(l)} \odot h'(\mathbf{Z}^{(l)})$ *Do not execute this line for last layer $l = L$*
  $d\mathbf{Q}^{(l-1)} = d\mathbf{Z}^{(l)}\mathbf{W}^{(l)}$
  $d\mathbf{W}^{(l)} = \frac{1}{n_b} d\mathbf{Z}^{(l)\mathsf{T}}\mathbf{Q}^{(l-1)}$
  $db_j^{(l)} = \frac{1}{n_b} \sum_{i=1}^{n_b} dZ_{ij}^{(l)} \qquad \forall j$
end
$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \left[\mathrm{vec}(d\mathbf{W}^{(1)})^{\mathsf{T}} \quad d\mathbf{b}^{(1)\mathsf{T}} \quad \cdots \quad \mathrm{vec}(d\mathbf{W}^{(L)})^{\mathsf{T}} \quad d\mathbf{b}^{(L)\mathsf{T}}\right]$
return $J(\boldsymbol{\theta})$, $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$

Figure 3

(question continues over)

(a) In Algorithm 6.1, what does $\mathbf{Q}^{(l)}$ represent? (2 marks)

(b) In Algorithm 6.1, what does n_b represent? (2 marks)

(c) In Algorithm 6.1, which line calculates the cross-entropy cost function? (2 marks)

(d) To train a neural network, Algorithm 6.1 is typically used as a subroutine of (stochastic) gradient descent. Explain how the information from Algorithm 6.1 (specifically, $\nabla_\theta J(\theta)$) is used to update the weights of the network. (4 marks)

(next question over)

Question 4.

(a) Compare the ensemble methods of bagging and boosting. Write a few sentences, giving one similarity and one difference between the two techniques. (5 marks)

(question continues over)

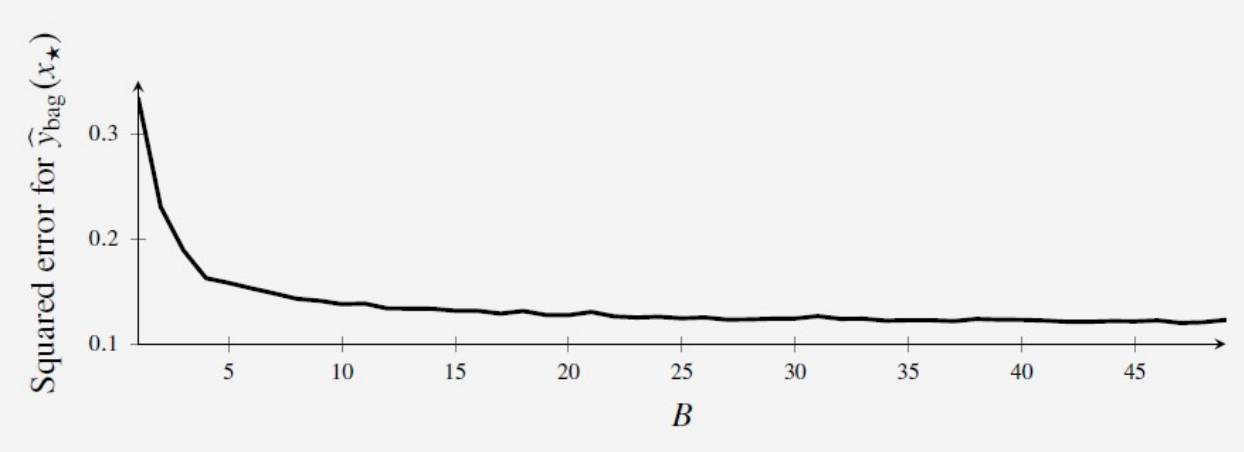

Figure 4

(b) Figure 4 is from the course textbook and shows an estimate of the generalisation error for a regression problem, as a function of the number of base models (B) in a bagged ensemble. Explain *what* the graph tells us and *why* this trend is to be expected, by referring to the bias-variance decomposition of error. (5 marks)

(next question over)

Question 5.

Figure 5 is taken for a machine learning textbook and shows some experimental results comparing neural networks trained with and without weight decay. The box-whiskers plots summarize the distribution of the error values obtained over repeated trials: each box shows (where visible) the median (stripe inside box), the range of the "middle 50%" of the data (box), and any result values beyond this as whiskers or individual point values.

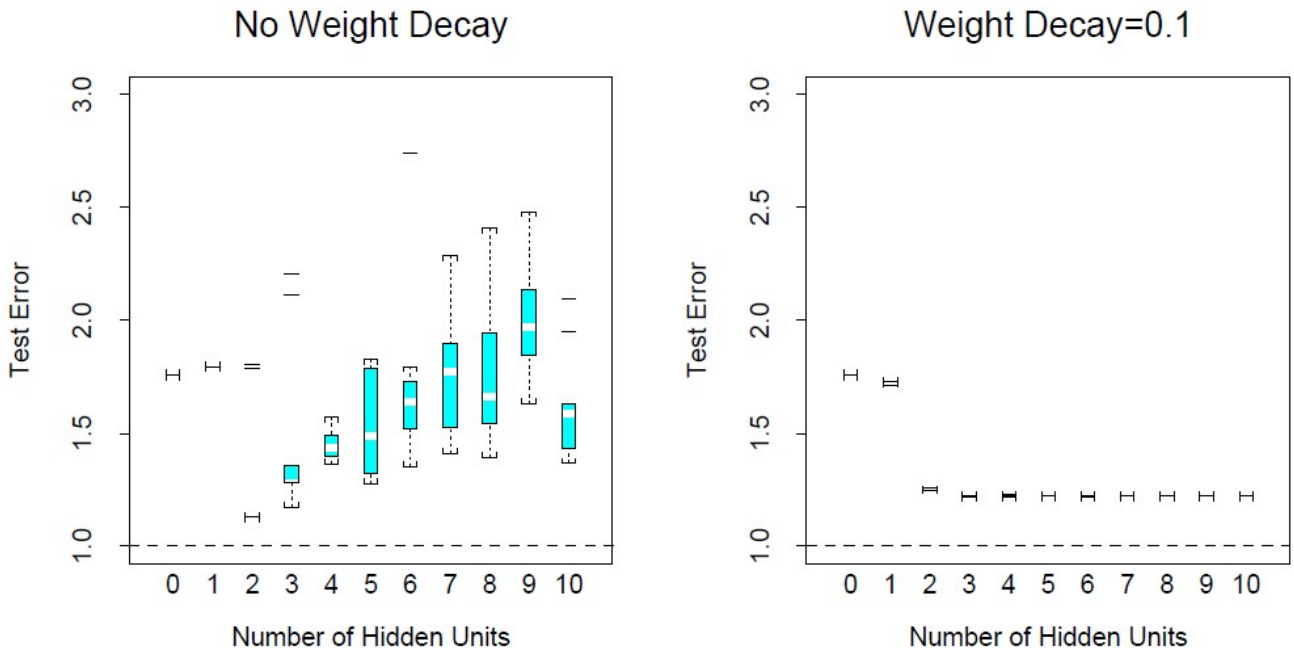

FIGURE 11.7. *Boxplots of test error, for simulated data example, relative to the Bayes error. True function is a sum of two sigmoids. The test error is displayed for ten different starting weights, for a single hidden layer neural network with the number units as indicated. The two panels represent no weight decay (left) and strong weight decay* $\lambda = 0.1$ *(right).*

Figure 5

(a) What do the results in Figure 5 show regarding the effect of weight decay? (3 marks)

(question continues over)

(b) What do the results in Figure 5 show regarding the effect of varying the number of hidden units? (4 marks)

(c) Consider a convolutional neural network where, at some point in the network:

- the input data is a 6×6 matrix
- convolution is applied with a 3×3 kernel, followed by max pooling using a 2×2 kernel.

Calculate the size of the output for these operations. (3 marks)

(next question over)

Question 6.

For both kernel ridge regression and support vector regression, the equation for predicting a text point is identical:

$$\hat{y}(\mathbf{x}_\star) = \hat{\alpha}^\top K(\mathbf{X}, \mathbf{x}_\star)$$

(a) Explain what is different about kernel ridge regression and support vector regression. You might like to discuss the size of the models, the computation involved or the training process. (4 marks)

(b) k-means is a well-known clustering algorithm. Explain briefly how it works (i.e. the two main steps involved) and how this can be seen as a simplified version of the expectation-maximization (EM) algorithm for fitting a Gaussian mixture model. (6 marks)

(next question over)

Question 7.

Figure 6 shows a Gaussian Process model for a one-dimensional regression problem. Figure 6(a) shows four samples from the prior distribution of the GP, and Figure 6(b) shows the posterior distribution after two data points have been observed, including the mean prediction (solid line) and four samples from the posterior (dashed lines). The shaded region indicates twice the standard deviation at each input value x.

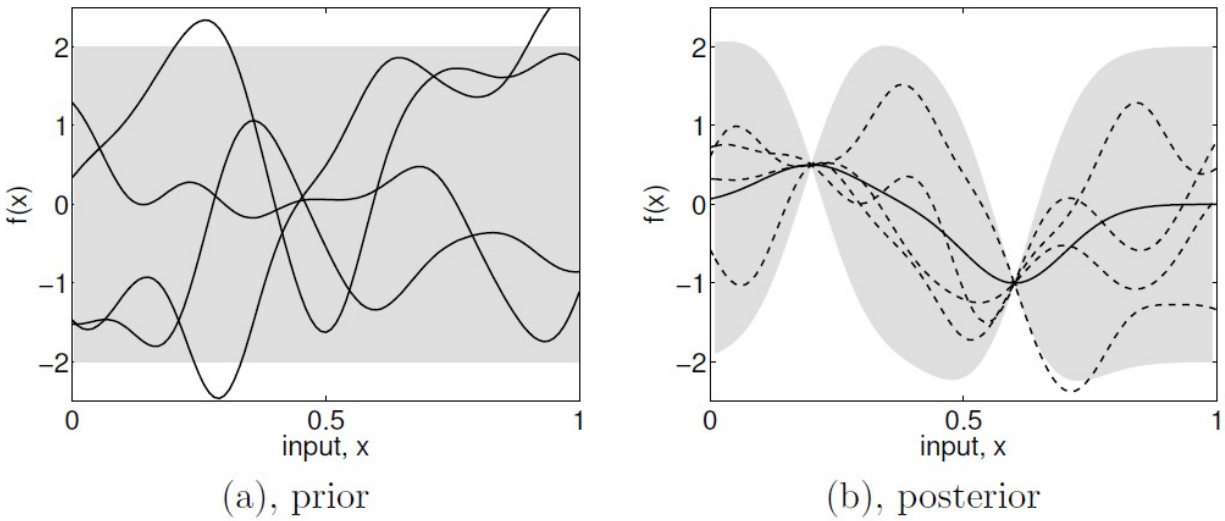

Figure 6

(a) Using Figure 6, explain the general principle of Bayesian inference in machine learning.(4 marks)

(question continues over)

(b) Explain the shape of the shaded region in Figure 6, and how and why it changes from Figure 6(a) to Figure 6(b). (3 marks)

(c) One of the advantages of GPs is their ability to provide a confidence value with its predictions. Using Figure 6(b), explain the general relationship between this value and the training data points. (3 marks)

(next question over)

Question 8.

1. Explain briefly what a *generative model* is in machine learning (i.e. what makes it a generative model as opposed to a non-generative model?). (4 marks)

2. Generative Adversarial Networks are trained using unlabelled data, but the training is formulated as two supervised learning problems and uses (stochastic) gradient descent. Explain the key steps of this training process. (4 marks)

3. Explain how the generator is used once training is complete? (2 marks)

END OF EXAMINATION

This page can be used if you need extra space to answer any of the questions. Clearly indicate the question(s) that you are answering. Ensure that you also put a note within the body of the relevant question(s) directing the marker here.